

SetProbe: Description-First Relation-Wise Self-Consistency for Closed-Book Knowledge Base Construction

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Abstract

The AKBC 2026 shared task asks a language model to recover the complete set of objects for a given subject–relation pair without retrieval, external knowledge bases, or additional training. We describe SetProbe, a relation-specific probing system that runs an open-weight language model locally. Each query receives a short identity description followed by multiple rationale-bearing generations with rotating training examples. Relation-wise aggregation uses numeric medians, generation-first empty-answer voting, and low-threshold set union. A targeted completion step is retained for borders and awards, while verification is restricted to the already strong border workflow. Deterministic parsing and train-derived alias normalization convert the generations into submission-format JSONL. Thresholds are calibrated only on complete training traces. This draft specifies the method and experimental protocol; quantitative results for the new system will be added after the replacement model is evaluated.

1 Introduction

How can knowledge stored in a language model be turned into a complete database entry? A model may readily answer that the Dominican Republic borders Haiti, yet the same question becomes considerably harder when the expected answer contains nine neighboring countries, 80 award recipients, or no object at all. The AKBC 2026 shared task isolates this problem: given a subject s and a relation r , a system must predict the complete set of object strings $\{o_1, \dots, o_k\}$ (AKBC Shared Task Organizers, 2026).

Consider a country with three gold-standard neighbors. Returning two correct neighbors yields perfect precision but only $2/3$ recall; adding three plausible but incorrect neighbors restores no recall and reduces precision to $2/5$. Empty sets introduce the opposite problem, since inventing a single death

Relation	Rows	Empty	Max. objects
Area	100	0	1
Capacity	100	0	1
Death city	100	39	1
Stock exchange	100	36	4
Land borders	68	18	14
Award recipients	10	0	610

Table 1: Cardinality of the AKBC 2026 validation split. “Empty” counts rows without a gold object.

city for a living person changes an otherwise perfect row into a complete miss. The central design question is therefore not merely whether the model knows a fact, but when another candidate should be admitted to the predicted set.

We address this question with description-first relation-wise self-consistency. The system first identifies the exact subject, then generates independent candidate sets through prompts that expose different recall paths, such as scale checks, clockwise border enumeration, or chronological award enumeration. Empty answers participate directly in aggregation instead of blocking generation through a preliminary gate. The main takeaway is simple: relations with different cardinalities and failure modes should share one orchestration pattern but use different aggregation rules.

2 Task

The dataset contains six relations and 477 training, 478 validation, and 477 test rows. Table 1 summarizes the validation split, which we reserve for evaluation; prompts use training examples and thresholds are fitted only on training traces. String predictions are matched after normalization and alias expansion. Numeric predictions for `hasArea` and `hasCapacity` are accepted within a relative tolerance of 5%. The official score is macro-averaged precision, recall, and F1 over subject–relation pairs.

The setting is closed-book: inference may use

one or more open-weight neural models with at most 32 billion total published parameters, but it may not query the web, a retrieval index, an external corpus, or a knowledge base. The rules also prohibit fine-tuning and continued pretraining. In-context examples, multi-step inference, and non-neural post-processing are allowed (AKBC Shared Task Organizers, 2026). Our system consequently uses the training split only for few-shot demonstrations and uses no external factual source.

3 System

3.1 Common inference procedure

All calls use one locally hosted open-weight model through an OpenAI-compatible LM Studio endpoint. For every subject–relation pair, the model first produces a short identity anchor describing entity type, scope, and likely namesake confusions. The same model then receives this anchor as explicitly uncertain context and generates n_r concise rationales and candidate sets. Each pass uses a different seed, recall focus, and rotating leave-one-subject-out sample of training demonstrations. For relations that permit empty answers, example selection alternates between empty and non-empty rows.

Candidate strings are parsed from the first valid JSON structure and normalized with the official evaluator’s punctuation, apostrophe, case, and diacritic rules. We additionally construct relation-specific alias maps from the nested aliases in the training split. An alias is rewritten only if all of its training occurrences identify the same canonical entity. Alias canonicalization happens before support is counted, so equivalent surfaces reinforce rather than split a vote. Raw responses, parsed values, rationales, seeds, token counts, and aggregation decisions are retained in immutable per-row traces.

We compare two output-control modes while holding model weights, prompts, and decoding settings fixed. The first requests JSON in natural-language instructions and applies a tolerant parser. The second converts Pydantic response classes into JSON Schema and asks LM Studio to constrain generation with the corresponding grammar. Schema constraints guarantee structure rather than factual correctness; if a constrained generation is interrupted or fails validation, the same tolerant parser is used as a recorded fallback. For awards, the schema separates recipient, work, and year before

aggregation.

3.2 Numeric relations

For area and capacity, respectively 20 and 15 prompts request independent estimates. The area prompts distinguish total surface area from land-only area and explicitly request square kilometres. The capacity prompts distinguish maximum attendance from a smaller seated configuration. Consider estimates of 35,000, 35,100, and 50,000 spectators: the system returns the median, 35,100, rather than allowing one recalled concert configuration to determine the prediction.

3.3 Death locations

Twenty passes independently recall the most specific death locality or return an empty candidate set. Empty is treated as a first-class candidate and wins only when it has a strict plurality and reaches a threshold calibrated from complete training traces. Otherwise the highest-support locality is retained, with at most one city in the final output. This generation-first rule avoids discarding a recoverable city because a separate status classifier was uncertain.

3.4 Stock exchanges

Ten passes separately recall primary listings, secondary or dual listings, and the legal entity attached to each exchange. Candidate exchanges survive when their pass-level support reaches a train-calibrated threshold. Empty may still win by strict plurality at a separate train-calibrated threshold. Prompts ask for exchange names rather than ticker symbols and explicitly distinguish a subsidiary from its listed parent.

3.5 Land borders

Border candidates are generated in three ways: clockwise enumeration, independent geographic recall, and a focused check for enclaves, exclaves, short borders, and integral overseas territories. A separate completion prompt sees the merged set and returns only omitted candidates. The verifier then removes maritime neighbors and borders that arise solely through non-integral dependencies. **[TODO: Evaluate a deterministic symmetry check using pairs that occur in the supplied splits.]**

3.6 Award recipients

A single output may contain 610 recipients, for which one undifferentiated prompt is prone to both truncation and omission. We therefore partition recall into five time windows—before 1960, 1960–1979, 1980–1999, 2000–2014, and 2015–1 July 2026—and add a sixth pass organized by profession, nationality, and recipient type. Twenty seeded passes cycle through these foci. A recipient is retained when it occurs in at least 5% of passes, which reduces to a high-recall union at $n = 20$. One completion pass searches for missing editions and recipient types. We apply no binary award verifier because preliminary experiments showed that it removed many correct long-tail recipients.

4 Experimental Setup

Models. We will compare [TODO: Gemma 3 27B-IT, Mistral Small 3.2 24B, and the completed GPT-OSS-20B reference run]. Each model is evaluated separately, so the inference-time parameter total remains at or below 32B. Quantization affects memory use but not the parameter count under the shared-task rules.

Inference configuration. The current configuration uses five in-context examples except for awards, where one large demonstration protects the context budget. It uses temperature 0.6, top- p 0.95, and a maximum generation length of 2,048 tokens. Relation sample counts are given above. [TODO: Report context length, LM Studio version, hardware, decoding parameters for every model, and total runtime.]

Model and threshold selection. We first compare models on a fixed 12-row stratified smoke set containing two examples per relation. All support and empty-answer thresholds are selected from complete leave-one-out training traces with the official evaluator. The locked configuration is then evaluated once on all 478 validation rows. We do not tune thresholds or entity-specific rules against validation answers.

Baselines and ablations. The primary historical baseline is the completed GPT-OSS-20B prototype. The new system removes its hard empty-answer gates, numeric selector, generic verifier, and award verifier. Controlled comparisons vary the eligible model and replace prompted JSON with schema-constrained output while keeping the pipeline fixed.

System	Macro P	Macro R	Macro F1
Official 9B baseline	.324	.507	.313
GPT-OSS-20B (legacy)	—	—	.460
SetProbe	—	—	—

Table 2: Validation results across all relations. The official baseline numbers are reported by the shared-task organizers.

Additional ablations vary self-consistency sample count and the optional train-derived SyntheticCoT demonstrations.

5 Results

Table 2 reserves the primary comparison. The final paper will additionally report per-relation scores, the number of empty predictions, average set size, inference time, and the ablation results. [TODO: Replace this paragraph with a quantitative account of where precision and recall change.]

6 Error Analysis

[TODO: Analyze at least 50 validation errors, separating missing model knowledge, identity errors, alias or parsing errors, incorrect abstention, incomplete enumeration, obsolete facts, and border-verification failures. Include one worked example for a null relation, one numeric relation, borders, and awards.]

7 Related Work

The shared task follows work that probes factual knowledge in pretrained language models, including LAMA (Petroni et al., 2019), but removes the assumption that each query has exactly one answer. [TODO: Add prior LM-KBC shared task papers, work on set-valued knowledge extraction, self-consistency, factual calibration, and closed-book question answering.]

8 Conclusion

We presented a single description-first self-consistency pipeline for closed-book knowledge base construction. It shares one orchestration pattern across relations while assigning numeric, empty-aware, union, and threshold aggregation according to relation cardinality. Train-only calibration and alias maps keep non-neural decisions within the challenge’s evidence boundary. The remaining empirical question is whether a more

knowledgeable eligible model supplies enough correct numeric and identity-level candidates to exceed the strong validation reference.

Limitations

The system cannot recover facts absent from the selected model’s pretraining data, which is particularly relevant for deaths, listings, and award recipients close to the July 2026 reference date. Reusing the same model for border generation and verification may reinforce rather than correct a misconception. Self-consistency stabilizes recalled candidates but cannot recover a fact that never appears in any sample, while award enumeration can remain incomplete after temporal partitioning. Train-calibrated thresholds may also shift under distribution change. We therefore preserve the test data for one final evaluation and record every prompt, response, configuration, and row shard.

References

- AKBC Shared Task Organizers. 2026. [AKBC Shared Task 2026: Predicting Complete Knowledge Base Entries from Language Models](#). Accessed 31 July 2026.
- Fabio Petroni, Tim Rocktäschel, Patrick Lewis, Anton Bakhtin, Yuxiang Wu, Alexander H. Miller, and Sebastian Riedel. 2019. [Language models as knowledge bases?](#) In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing*, pages 2463–2473. Association for Computational Linguistics.

A Prompt and Configuration Inventory

[TODO: Include the final prompt templates, parsing rules, random seeds, and relation-specific thresholds required to reproduce the submitted run.]